

Deployment Guide

This guide explains how to deploy DriveAbility to your own environment. The primary deployment target is **Vercel**, but Docker Compose configurations are also provided for self-hosting on your own infrastructure.

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Architecture Recap

Browser ↔ Next.js (client/) ↔ Express API (server/) ↔ MongoDB Atlas

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Vercel serverless (api/index.ts) — production only

- **Frontend**: Next.js 14 (Pages Router), built and served by Vercel (or Next.js `start` in Docker).
- **Backend**: Express.js, TypeScript. In production on Vercel, the entire Express app is bundled into a single serverless function at `api/index.ts`. All `/api/*` requests are rewritten to that function via `vercel.json`.
- **Database**: MongoDB (Atlas in production, or a local container for self-hosting).
- **Media**: Uploaded images are stored as base64 data URIs in MongoDB — no external object storage is needed.

Prerequisites

Tool	Version	When Needed
Node.js	v18+	Always (Vercel uses this automatically)
npm	v9+	Always
Git	Any recent	Always
Vercel CLI (optional)	Latest	If deploying from the command line
Docker & Docker Compose	v20+ / v2+	Only for self-hosted deployments
MongoDB Atlas account	—	Vercel deployments (or any deployment without a local MongoDB)

Option 1 — Deploy to Vercel (Recommended)

Vercel is the simplest path. The repo includes a `vercel.json` that handles the build, output, and API rewrites automatically.

Step 1: Fork or Clone the Repository

```
```bash
git clone
cd mc-website
```

## Step 2: Create a Vercel Project

### Via the Vercel Dashboard (easiest):

1. Go to [vercel.com/new](https://vercel.com/new)
2. Import your GitHub/GitLab/Bitbucket repository
3. Vercel will auto-detect the Next.js framework — no overrides needed
4. Do not change the build settings; vercel.json handles everything:
  1. **Install Command:** `npm install && cd client && npm install --include=dev && cd ../server && npm install --include=dev`
  2. **Build Command:** `cd client && npm install --include=dev && npm run build`
  3. **Output Directory:** `client/.next`

### Via the Vercel CLI:

```
npm i -g vercel
vercel login
vercel # follow the prompts to link to a project
```

## Step 3: Configure Environment Variables

In the Vercel project dashboard, go to **Settings** → **Environment Variables** and add:

Variable	Value	Required
MONGODB_URI	Your MongoDB Atlas connection string	✓
JWT_SECRET	A 32+ character random string (openssl rand -base64 32)	✓
JWT_EXPIRE	30d (or your preferred token lifetime)	✓
NODE_ENV	production	✓
FRONTEND_URL	https://your-app.vercel.app (or your custom domain)	✓
EMAIL_HOST	SMTP hostname (e.g., smtp.mailtrap.io)	✓ for email
EMAIL_PORT	587	✓ for email
EMAIL_SECURE	false (use true for port 465)	✓ for email
EMAIL_USER	SMTP username	✓ for email
EMAIL_PASSWORD	SMTP password	✓ for email
EMAIL_FROM	DriveAbility	✓ for email
NEXT_PUBLIC_GA_MEASUREMENT_ID	Your Google Analytics 4 Measurement ID (e.g., G-XXXXXXXXXX)	Optional

⚠ Critical: The database variable must be named `MONGODB_URI` — not `MONGO_URI`. The server code reads `process.env.MONGODB_URI` exclusively.

**Tip:** Set variables for all environments (Production, Preview, Development) or scope them appropriately. Preview deployments from non-production branches will use Preview-scoped variables.

# Step 4: Deploy

## Automatic (recommended):

Connect the release branch as the production branch in Vercel. Every push to release triggers a production deployment.

main    ✖ development work, PRs merge here  
release ✖ Vercel production deployment target

Workflow:

- 1. Branch from main for features
- 2. Merge PRs into main
- 3. When ready, merge main ✖ release to deploy

## Manual:

vercel --prod

# Step 5: Create an Admin Account

- 1. Register a user at <https://your-app.vercel.app/register>
- 2. Promote the user to admin via MongoDB Atlas shell or Compass:

```
db.users.updateOne(
 { email: "you@example.com" },
 { $set: { role: "admin" } }
)
```

- 3. Access the admin dashboard at /admin

# How Vercel Serves the App

Component	How It Works
Frontend	Vercel builds the Next.js app from client/ and serves pages as static + SSR
Backend API	The Express app is wrapped in api/index.ts and deployed as a single serverless function
Routing	vercel.json rewrites all /api/* requests to the serverless function

DB Connection	api/index.ts uses a module-level isConnected flag to reuse the MongoDB connection across warm invocations
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## Option 2 – Self-Host with Docker

Docker Compose files are provided for both development and production. This is ideal for deploying on a VPS, Raspberry Pi, or any machine where you want full control.

### Required Software

Tool	Version	Purpose
Docker	v20+	Container runtime
Docker Compose	v2+	Multi-container orchestration
Git	Any	Clone the repo

No Node.js installation is needed on the host – everything runs inside containers.

### Development (with Hot Reload)

`docker compose up --build`

This starts three containers:

Container	Port	Description
assistive-tech-mongo	27017	MongoDB 7 with persistent volume
assistive-reviews-main	5001	Express backend with nodemon (auto-restart on changes)
assistive-tech-client	3000	Next.js dev server

Source directories are bind-mounted, so edits on your host are reflected immediately.

### Production

```
docker compose -f docker-compose.prod.yml up --build -d
```

This starts four containers:

Container	Exposed Port	Description
assistive-tech-mongodb-prod	None (internal only)	MongoDB 7, data persisted in a named volume
assistive-tech-server-prod	None (internal only)	Express production build
assistive-tech-client-prod	None (internal only)	Next.js production build
assistive-tech-nginx	80, 443	Nginx reverse proxy routing traffic to client & server

Before running, create a .env file at the repo root with production secrets:

```
JWT_SECRET=
JWT_EXPIRE=30d
```

The production compose file reads JWT\_SECRET and JWT\_EXPIRE from the host environment (or .env).

## Nginx Configuration

The production stack includes an Nginx reverse proxy. Edit nginx/default.conf to:

- Configure your domain name
- Set up SSL certificates (mount them into nginx/ssl/)
- Adjust upstream routing as needed

## Docker vs Vercel: When to Choose What

	Vercel	Docker Self-Host
Setup effort	Minimal – connect repo and go	Moderate – need a server, DNS, SSL
Scaling	Automatic (serverless)	Manual (scale containers yourself)
Cost	Free tier available	Cost of your own hosting
Database	Must use MongoDB Atlas (external)	Can run MongoDB in a container

Media uploads	Stored as base64 in MongoDB	Same
Custom domain	Supported via Vercel dashboard	Configure in Nginx
SSH/full control	No	Yes

## Environment Variables Reference

All variables are set at the **repo root** – there is no separate `server/.env`. In Vercel, set them in the project dashboard.

### Required

Variable	Example	Description
MONGODB_URI	mongodb+srv:// user:pass@cluster.mongodb.net/db	MongoDB connection string. <b>Must be MONGODB_URI</b> , not MONGO_URI
JWT_SECRET	<32+ char random string>	Secret for signing JWTs. Generate with <code>openssl rand -base64 32</code>
JWT_EXPIRE	30d	JWT token lifetime (e.g., 30d, 7d, 1h)
NODE_ENV	production	development   production   test

### Recommended for Production

Variable	Example	Description
FRONTEND_URL	https://your-app.vercel.app	Used for CORS allowlist and password reset email links. Falls back to VERCEL_URL if unset
PORT	5001	Express server port (Docker only; Vercel ignores this)

### Email (Required in Production)

Variable	Example	Description
EMAIL_HOST	smtp.mailtrap.io	SMTP server hostname

EMAIL_PORT	587	SMTP port (587 for STARTTLS, 465 for SSL)
EMAIL_SECURE	false	true for port 465, false for STARTTLS
EMAIL_USER	your_smtp_user	SMTP username
EMAIL_PASSWORD	your_smtp_pass	SMTP password
EMAIL_FROM	DriveAbility	Sender address for outgoing emails

In development, if email variables are not set, the server auto-creates an [Ethereal](#) test account and prints a preview URL to the console.

## Analytics (Optional)

Variable	Example	Description
NEXT_PUBLIC_GA_MEASUREMENT_ID	G-Y3952ZJ8ZE	Google Analytics 4 Measurement ID. Falls back to a hardcoded default if not set

## External Resources & Services

### MongoDB Atlas – Database

	Details
What it does	Hosts the MongoDB database (users, products, reviews, profiles, settings, feedback, resources)
Why it's needed	The app requires MongoDB. Vercel has no built-in database; Atlas is the standard cloud option
Free tier	<b>M0 Shared Cluster</b> – 512 MB storage, shared RAM, free forever. Sufficient for development and small deployments
Paid tiers	M2 ( $\frac{9}{mo} \rightarrow M10 + (\frac{57}{mo+})$ ) for dedicated resources, backups, and higher storage
Sign up	<a href="https://mongodb.com/atlas">mongodb.com/atlas</a>
Setup notes	Create a cluster, add a database user, whitelist IPs (Vercel requires 0.0.0.0/0 since serverless functions use dynamic IPs), and copy the connection string



## Vercel – Hosting & Serverless

	Details
What it does	Hosts the Next.js frontend and the Express backend as a serverless function
Free tier	Hobby – 100 GB bandwidth/mo, 100 GB-hours serverless execution, 1 concurrent build. Suitable for personal projects and demos
Paid tiers	Pro (\$20/user/mo) for team features, higher limits, and preview deployments at scale
Sign up	<a href="https://vercel.com">vercel.com</a>

## Vercel Analytics – Web Analytics

	Details
What it does	Provides pageview and custom event tracking directly in the Vercel dashboard via the @vercel/analytics package
Free tier	Included with all Vercel plans – up to 2,500 events/month on Hobby
Paid tiers	Pro includes more events; additional events billed at \$0.65/1,000
Setup	Automatic – the `` component is already wired into _app.tsx. Custom events use track() from @vercel/analytics

## Google Analytics 4 – Extended Analytics

	Details
What it does	Provides detailed user behavior analytics, audience insights, and conversion tracking
Free tier	Fully free – GA4 has no paid tier for standard usage
Setup	The GA4 tag is loaded via next/Script in _app.tsx. Set NEXT_PUBLIC_GA_MEASUREMENT_ID to your own Measurement ID, or the app falls back to a default. Custom events are fired via client/src/utis/analytics.ts using window.gtag()
Sign up	<a href="https://analytics.google.com">analytics.google.com</a>

# SMTP Email Provider — Transactional Email

	Details
What it does	Sends password reset emails and email verification messages
Why it's needed	Required in production for password reset and email verification flows
Recommended providers	

Provider	Free Tier	Paid Tier	Notes
<a href="#">Mailtrap</a>	✓ 1,000 emails/mo	From \$15/mo	Purpose-built for dev testing; also has a sending API
<a href="#">Brevo (Sendinblue)</a>	✓ 300 emails/day	From \$9/mo	Full marketing suite included
<a href="#">Mailgun</a>	✓ 100 emails/day (first 3 months)	From \$15/mo	Developer-friendly API and SMTP
<a href="#">SendGrid</a>	✓ 100 emails/day forever	From \$19.95/mo	Owned by Twilio; generous free tier
<a href="#">Amazon SES</a>	✓ 3,000 messages/mo (from EC2)	\$0.10 per 1,000 emails	Cheapest at scale

In **development**, you don't need any SMTP provider. The server auto-creates an [Ethereal](#) test account and logs preview URLs to the console.

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## Post-Deployment Checklist

- ☐ Environment variables — All required vars are set in Vercel dashboard (or .env for Docker)
- ☐ MongoDB Atlas IP whitelist — 0.0.0.0/0 is whitelisted for Vercel (serverless IPs are dynamic)
- ☐ Test the API — Visit <https://your-app.vercel.app/api> — should return { "message": "DriveAbility API is running" }
- ☐ Test authentication — Register a new user and verify login works
- ☐ Test email — Trigger a password reset and confirm the email arrives
- ☐

Create admin account — Register, then promote via MongoDB shell

- ☐ Verify CORS — FRONTEND\_URL matches your actual deployment URL; API calls from the frontend should not be blocked
  - ☐ Custom domain (if applicable) — Add in Vercel dashboard » Settings » Domains; update FRONTEND\_URL to match
  - ☐ Google Analytics — Verify events are arriving in the GA4 dashboard
  - ☐ Vercel Analytics — Check the Analytics tab in your Vercel project dashboard
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## Troubleshooting

### Deployment Fails on Vercel

- **Build error — missing dependencies:** Ensure vercel.json install command includes --include=dev for both client/ and server/. This is already configured in the repo's vercel.json.
- **TypeScript errors:** Run cd client && npx tsc --noEmit and cd server && npm run build locally to catch errors before pushing.

### API Returns 500 or "Cannot connect to MongoDB"

- Verify MONGODB\_URI is set correctly in Vercel environment variables (not MONGO\_URI)
- Check MongoDB Atlas IP whitelist — Vercel serverless functions use unpredictable IPs, so 0.0.0.0/0 must be allowed
- Confirm the database user password does not contain unescaped special characters

### CORS Errors in Browser Console

- FRONTEND\_URL must exactly match the URL in the browser bar (including https://)
- If using a custom domain, update FRONTEND\_URL to that domain
- The server also falls back to VERCEL\_URL (auto-set by Vercel) if FRONTEND\_URL is not configured

### Emails Not Sending in Production

- All EMAIL\_\* variables must be set — the server throws a startup error if they are missing in production mode

- Test your SMTP credentials independently (e.g., with a tool like swaks or a test script)

## Serverless Function Timeout

- Vercel Hobby has a 10-second function timeout (Pro: 60 seconds)
- Cold starts include MongoDB connection time — the isConnected flag in api/index.ts caches the connection for subsequent warm invocations
- If timeouts persist, check MongoDB Atlas cluster region — choose one close to your Vercel deployment region

## "JWT\_SECRET is not set" on Server Start

- This is a startup guard in server/src/index.ts — ensure JWT\_SECRET is set in your environment
- In Docker, check the .env file or docker-compose.prod.yml environment section